

Scenario 1: Online Course Completion Time Prediction

Predict how many **days** a student will take to complete an online course.

here we are taking 3 inputs

Inputs (3 features)

1. hours_per_day
2. previous_experience_years
3. number_of_modules

Output

- completion_days
- here's the code

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

# Dataset
data = {
    "hours_per_day": [1, 2, 3, 4, 5, 2, 3, 4, 5, 6],
    "experience_years": [0, 1, 2, 1, 3, 0, 2, 2, 4, 3],
    "modules": [10, 9, 8, 7, 6, 10, 8, 7, 5, 6],
    "completion_days": [60, 50, 40, 35, 25, 55, 42, 32, 22, 28]
}

df = pd.DataFrame(data)

# Features and Target
X = df[["hours_per_day", "experience_years", "modules"]]
y = df["completion_days"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=42)

# Model
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
```

```

new_student = [[4, 2, 7]]
prediction = model.predict(new_student)
print(f"Predicted course completion days: {prediction[0]:.0f}")

# VISUALIZATION

fig, axes = plt.subplots(1, 3, figsize=(15, 5))

axes[0].scatter(df["hours_per_day"], df["completion_days"], color="blue",
s=80)
axes[0].set_xlabel("Hours per Day")
axes[0].set_ylabel("Completion Days")
axes[0].set_title("Hours per Day vs Completion Days")

axes[1].scatter(df["experience_years"], df["completion_days"], color="green",
s=80)
axes[1].set_xlabel("Experience (Years)")
axes[1].set_ylabel("Completion Days")
axes[1].set_title("Experience vs Completion Days")

axes[2].scatter(df["modules"], df["completion_days"], color="red", s=80)
axes[2].set_xlabel("Number of Modules")
axes[2].set_ylabel("Completion Days")
axes[2].set_title("Modules vs Completion Days")

plt.tight_layout()
plt.show()

```

Scenario 2: Mobile Battery Drain Prediction

Problem

Predict **battery drain (%)** based on:

1. Screen time (hours)
2. Number of apps used
3. Brightness level (%)

```

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

```

```
data = {
    "screen_time": [1,2,3,4,5,6,2,3,5,4],
    "apps_used": [3,5,7,9,12,15,6,8,10,11],
    "brightness": [30,40,50,60,70,80,45,55,65,75],
    "battery_drain": [10,20,35,50,65,80,25,40,60,55]
}

df = pd.DataFrame(data)

X = df[["screen_time", "apps_used", "brightness"]]
y = df["battery_drain"]

model = LinearRegression()
model.fit(X, y)

new_usage = pd.DataFrame([[4,9,60]], columns=X.columns)
prediction = model.predict(new_usage)

print(f"Predicted battery drain: {prediction[0]:.1f}%")

# Visualization
plt.scatter(df["screen_time"], y)
plt.xlabel("Screen Time (hrs)")
plt.ylabel("Battery Drain (%)")
plt.title("Screen Time vs Battery Drain")
plt.show()
```